

# FusionCore — Executive Summary

## Three-Version Architecture Evaluation on NASA C-MAPSS Turbofan Corpus

FusionCore Programme — v0 through v2 — Final Report

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## Executive Summary

The FusionCore programme evaluated three progressively sophisticated predictive maintenance architectures on the NASA Commercial Modular Aero-Propulsion System Simulation corpus. The programme objective throughout was to surpass the v0 XGBoost Remaining Useful Life estimator on Critical-band recall whilst maintaining safety-gate compliance. After three versions spanning a tabular ensemble baseline, a physics-informed hybrid temporal network, and a calibrated uncertainty-quantified extension of that network, the objective was not achieved on the C-MAPSS corpus.

The v0 XGBoost baseline achieves a Critical-band recall of 0.9363 on the NASA Official Test Set, a Root Mean Square Error of 14.85 cycles and a projected net fleet saving of approximately USD 13.8 million per annum per 100-aircraft fleet. No subsequent PiNet variant surpasses this result on any safety-critical metric. The v1 Stage A Critical-band recall collapse to 0.4204 was substantially recovered by the v2 Stage B retraining to 0.7834, but the deficit against v0 remains 0.1529 units of recall, far beyond the permitted gate tolerance of 0.02.

The programme concludes with a recommendation to migrate the PiNet architecture to the NASA N-CMAPSS real-flight dataset. The architectural diagnosis from v1 and v2 is clear: PiNet's regime branch earns its keep on multi-regime data but cannot compensate for the absence of fault-mode disambiguation on single-regime dual-fault profiles. N-CMAPSS, with continuous flight-envelope variation and higher sensor noise, is the dataset where the physics-informed temporal hypothesis should next be tested.

**Table 1 — Programme Trajectory**

| Version | Scope                                                                                                    | Primary comparator             | Safety Gate                   | Best Critical Recall |
|---------|----------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------------|----------------------|
| v0      | EDA → zero-leakage normalisation → 91-feature engineering → XGBoost + neural SOTA benchmarks             | XGBoost vs TFT, N-HiTS, DeepAR | Reference                     | 0.9363 (XGBoost)     |
| v1      | Physics-informed hybrid TCN + physics network (PiNet) — Stage A training                                 | v0 XGBoost (carry-forward)     | FAIL ( $\Delta R_C = -0.51$ ) | 0.4204               |
| v2      | Stage B retraining under four mitigations + split conformal intervals + temperature-scaled probabilities | v0 XGBoost + v1 Stage A        | FAIL ( $\Delta R_C = -0.15$ ) | 0.7834               |

## 1. Headline Results Across All Model Variants

Table 2 presents the complete comparative result on the NASA Official Test Set of 707 engines. The Critical-Recall Safety Gate, defined as Equation 1, requires that any candidate model's Critical-band recall not fall more than two percentage points below the v0 XGBoost reference. No neural architecture passes this gate.

$$\Delta R_C = R_C^{\text{model}} - R_C^{\text{v0}} \geq -0.02 \quad [\text{Eq. 1}]$$

What the Safety Gate requires. The gate is the programme's binding acceptance test. It asks a single operational question: is the candidate model at least as good as the established baseline at identifying engines that need urgent attention? A recall deficit of no more than two percentage points is the permitted tolerance. Every PiNet variant tested to date has failed this gate; the v0 XGBoost baseline remains the reference.

**Table 2 — Headline Metrics — NASA Official Test Set (707 Engines)**

| Metric                | v0 XGBoost | v0 TFT     | v0 N-HITS  | v1 Stage A | v2 Stage B |
|-----------------------|------------|------------|------------|------------|------------|
| RMSE (cycles)         | 14.85      | ≈ 62       | ≈ 62       | 24.66      | 28.05      |
| MAE (cycles)          | 10.41      | ≈ 49       | ≈ 49       | 18.83      | 20.57      |
| NASA Asymmetric Score | 4,336      | 10,060,000 | 10,060,000 | 32,204     | 112,334    |
| Critical Recall $R_C$ | 0.9363     | 0.0446     | 0.0446     | 0.4204     | 0.7834     |
| Critical $F_2$        | 0.9339     | 0.0532     | 0.0532     | 0.4735     | 0.7716     |
| Critical Precision    | 0.9245     | 0.2333     | 0.2333     | 0.9565     | 0.7283     |
| Safety Gate           | Reference  | FAIL       | FAIL       | FAIL       | FAIL       |

The SOTA neural benchmarks (TFT and N-HITS) are catastrophically worse than XGBoost on the NASA test set: Critical-band recall below 0.05 and a NASA score nine orders of magnitude larger. Both disqualify entirely in the v0 four-quadrant evaluation framework<sup>1</sup>. PiNet in both v1 and v2 is at least a positive contributor — net fleet savings are positive at both versions — but neither closes the gap with v0.

How to read the numbers in Table 2. RMSE is the average prediction error in flight cycles; one cycle is approximately one flight. The NASA Asymmetric Score penalises optimistic predictions more heavily than cautious ones: a 30-cycle late error costs nearly twice as much as a 30-cycle early error, because the operational consequence of missing a failure is more severe than scheduling maintenance early. Critical-Band Recall is the fraction of engines genuinely approaching failure that the model correctly identifies; 0.9363 means 93.6 in every 100 critical engines are correctly flagged. The SOTA benchmarks (TFT, N-HITS) are net cash-negative: their false-alert volume at sub-0.05 Critical recall overwhelms any UER avoidance benefit.

<sup>1</sup>The v0 four-quadrant evaluation classifies models into: Quadrant A (RMSE good, NASA good), B (RMSE good, NASA poor), C (RMSE poor, NASA good), D (RMSE poor, NASA poor). TFT and N-HITS land in Quadrant B (low RMSE, catastrophic NASA score). XGBoost is the sole Quadrant A occupant.

## 2. Cross-Version Engineering Lessons

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### Lesson 1 — Sampler choice has out-of-distribution consequences

The v1 Stage A training sampler over-represents the Critical band at approximately 27.5 per cent of every minibatch, against its natural frequency of 16 per cent in the corpus. The model duly learns a decision policy calibrated to this inflated Critical frequency. At evaluation time, where the natural frequency applies, this produces systematic late-prediction bias: the model over-predicts Remaining Useful Life because it was trained to expect more Critical-band observations per minibatch than the test set delivers. The v2 Stage B correction — using the natural distribution plus class-weighted cross-entropy to preserve the per-class gradient signal — recovered 0.36 units of Critical recall.

### Lesson 2 — Post-hoc calibration does not transfer across distributional boundaries

Split conformal prediction intervals and temperature scaling both succeed on internal validation data and both fail on the NASA test set. The conformal intervals are exact to four decimal places on the calibration slice and 18–23 percentage points under-nominal on test. Temperature scaling reduces Expected Calibration Error by 71 per cent on validation and by only 12 per cent on test. The mechanism is the same in both cases: the NASA test set contains truncated trajectories whereas the internal validation set contains run-to-failure trajectories. The two sets do not satisfy the exchangeability condition that both methods require, and the calibration phase emits no statistical warning of this before the test set is opened<sup>2</sup>.

### Lesson 3 — Internal validation metrics do not predict test performance

At every version the validation metrics substantially outperform the corresponding test metrics. v1 Stage A validation Critical recall was 0.918; test was 0.420. v2 Stage B validation Critical recall was 0.873; test was 0.783. The implication is direct: any acceptance gate placed on validation-distribution metrics cannot verify whether the model will pass the corresponding test-distribution threshold. Programme gate design for v3 and beyond must account for this structural gap, either by cross-validating on partitions that reflect the test distribution or by applying explicit test-distribution similarity checks before using validation performance as an acceptance criterion.

### Lesson 4 — Dual-fault subsets are the architectural weakness

Across both v1 and v2, the single-regime dual-fault subset accounts for the dominant share of pooled regression and classification error. In v2 this subset contributes 83.6 per cent of the pooled NASA score from only 14 per cent of the test engines and delivers a Critical-band recall of 0.40 against 0.98 on the multi-regime single-fault subset. The architectural explanation is unintuitive: PiNet's regime branch provides strong discriminative context when operational conditions vary across regimes, as on the multi-regime subsets. When all engines operate in a single regime and the only discriminative signal is the sensor-signature evolution under two competing fault modes, the regime branch contributes nothing, and the temporal branch must resolve ambiguity that the current feature representation does not adequately encode.

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<sup>2</sup>This is a known limitation of standard split conformal and temperature scaling, not an implementation defect. Distribution-aware alternatives — conformalised quantile regression (Romano et al. 2019), localised conformal prediction (Guan 2023), or weighted exchangeability corrections (Tibshirani et al. 2019) — would be appropriate for v3.

**Table 3 — v2 Per-Subset Result: Regime versus Fault-Mode**

| Subset | Regime | Fault mode | RMSE | NASA Score | Critical Recall $R_c$ | Share of pooled NASA |
|--------|--------|------------|------|------------|-----------------------|----------------------|
| FD001  | Single | Single     | 23.3 | 3,553      | 0.640                 | 3.2%                 |
| FD002  | Multi  | Single     | 24.0 | 2,600      | 0.983                 | 2.3%                 |
| FD003  | Single | Dual       | 44.1 | 93,873     | 0.400                 | 83.6%                |
| FD004  | Multi  | Dual       | 25.2 | 12,308     | 0.769                 | 11.0%                |

### Lesson 5 — TCN temporal capacity exceeds what C-MAPSS demands

The v0 XGBoost baseline operates on instantaneous features with no sliding window or temporal context. PiNet’s temporal branch uses a thirty-cycle dilated convolutional stack designed to exploit the long-horizon autocorrelation structure identified in the v0 exploratory analysis — notably the autocorrelation function on the exhaust-gas temperature sensor, which shows significant lags extending to cycle seventeen. Despite this architectural advantage, v2 PiNet achieves a Critical-band recall of 0.78 against v0 XGBoost’s 0.94<sup>3</sup>. The C-MAPSS simulator’s signal-to-noise ratio is high enough that a tabular gradient-boosted model captures the relevant structure without a temporal window.

### Lesson 6 — Confounded experimental changes are unrecoverable

The v1 Stage A training configuration departed from accepted predictive maintenance practice on four parameters simultaneously: an excessively high learning rate, insufficient regularisation, a distribution-distorting training sampler and an imbalanced gradient budget between the regression and classification objectives. The v2 Stage B correction applied all four mitigations simultaneously, recovering 0.36 units of Critical recall, but cannot attribute the recovery to any individual change. A disciplined experimental protocol that introduced each change in a separate ablation would have produced a more informative causal decomposition at the cost of additional training runs. For v3, any multi-parameter departure from the v2 Stage B canonical configuration should be ablated one parameter at a time.

## 3. Financial Implications

### 3.1 The Net Savings Model

The v0 programme Phase 5 financial analysis produces fleet-level net savings estimates from two primary quantities: the annual avoided Unscheduled Engine Removal (UER) saving, which is proportional to Critical-band recall, and the annual false-alert cost, which is proportional to one minus Critical-band precision scaled by the positive-prediction rate. Using a cost of USD 1.5 million per UER and USD 0.30 million per false alert, the model takes the form given in Equation 2<sup>4</sup>.

$$Net = R_c \cdot N_{UER} \cdot C_{UER} - (1 - P_c) \cdot N_{FA} \cdot C_{FA} \quad [\text{Eq. 2}]$$

In this expression,  $R_c$  is Critical-band recall,  $P_c$  is Critical-band precision,  $N_{UER}$  is the total number of actual UERs per fleet per annum,  $N_{FA}$  is the base false-alert volume under the natural positive-prediction rate, and  $C_{UER}$  and  $C_{FA}$  are the respective unit costs. The model captures the fundamental asymmetry

<sup>3</sup>This does not mean the temporal architecture is incorrectly designed. The C-MAPSS simulator generates relatively clean, high signal-to-noise trajectories where the instantaneous feature snapshot is sufficient for accurate RUL estimation. N-CMAPSS, with higher sensor noise and continuous operational variability, is the appropriate test environment for the temporal denoising hypothesis.

<sup>4</sup>The USD 1.5M per UER and USD 0.30M per false alert are illustrative industry-benchmark figures used for comparative analysis. They are derived from published airline maintenance cost studies. All v0 Phase 5 financial figures are computed transparently from these parameters and should not be treated as contractual estimates.

of the decision problem: high recall avoids expensive UERs, but low precision generates false alerts that carry their own AOG-day cost.

### 3.2 v0 Baseline and SOTA Benchmarks

Note on the financial figures. The USD 1.5 million per UER and USD 0.30 million per false alert are illustrative industry-benchmark figures derived from published airline maintenance-economics studies. They are used consistently across all model variants to enable direct like-for-like comparison. They should not be treated as contractual estimates for any specific fleet or operator.

**Table 4 — v0 Financial Outcome per 100-Aircraft Fleet per Annum**

| Model      | UERs avoided | UER savings | False alerts | FA cost   | Net savings |
|------------|--------------|-------------|--------------|-----------|-------------|
| v0 XGBoost | 9.36         | USD 14.04M  | 0.75         | USD 0.22M | USD 13.82M  |
| v0 TFT     | 0.45         | USD 0.67M   | 7.67         | USD 2.30M | –USD 1.63M  |
| v0 N-HITS  | 0.45         | USD 0.67M   | 7.67         | USD 2.30M | –USD 1.63M  |

The SOTA neural benchmarks are net cash-negative: their false-alert volume at sub-0.05 Critical recall translates to more false-alert cost than UER-avoidance value. This is the mechanism by which the v0 four-quadrant evaluation framework disqualified them irrespective of their raw RMSE: a model that generates 7.67 false alerts per annum per 100 aircraft at USD 0.30 million each is not viable regardless of its cycle-level prediction accuracy.

### 3.3 PiNet Extrapolated Financials

The v0 financial model is applied to the v1 and v2 recall and precision values. The extrapolation assumes a constant fleet size of 100 aircraft and the same unit cost parameters, producing a consistent like-for-like comparison.

**Table 5 — Extrapolated Financial Comparison Across All PiNet Variants**

| Model      | $R_c$  | $P_c$  | UERs avoided | UER savings | False alerts | FA cost   | Net savings | $\Delta$ vs v0 |
|------------|--------|--------|--------------|-------------|--------------|-----------|-------------|----------------|
| v0 XGBoost | 0.9363 | 0.9245 | 9.36         | USD 14.04M  | 0.75         | USD 0.22M | USD 13.82M  | —              |
| v1 Stage A | 0.4204 | 0.9565 | 4.20         | USD 6.31M   | 0.19         | USD 0.06M | USD 6.25M   | –USD 7.57M     |
| v2 Stage B | 0.7834 | 0.7283 | 7.83         | USD 11.75M  | 2.91         | USD 0.87M | USD 10.88M  | –USD 2.94M     |

v2 recovers approximately USD 4.6 million per annum per 100-aircraft fleet relative to v1 Stage A, primarily through the recall recovery. The remaining USD 2.9 million annual gap against v0 XGBoost is driven almost entirely by v2's lower Critical precision (0.728 versus 0.924), which generates approximately four times the false-alert volume. Scaled to a 200-aircraft fleet the v2 vs v0 deficit is approximately USD 5.9 million per year; at a 50-aircraft fleet it is approximately USD 1.5 million.

*The precision lever, not the recall lever, is what closes the financial gap with v0 XGBoost. v2 has already achieved 0.78 recall; the financial shortfall is a precision problem. v3 design should therefore treat Critical precision as the primary financial metric. If v3 can restore precision to approximately 0.92 whilst preserving v2's recall, the financial model projects net savings of USD 11.55 million per 100-aircraft fleet — USD 2.3 million short of v0 — and a further recall improvement to 0.94 closes the gap to within USD 0.03 million.*

What the v3 scenarios show. The two rows project what the financial outcome would be if v3 achieves a specific combination of recall and precision. The key insight is that the precision lever — not the recall lever — closes the financial gap with v0 XGBoost. v2 already achieves a recall of 0.78; the remaining USD 2.9 million annual deficit is almost entirely attributable to v2's lower precision generating approximately four times the false-alert volume of v0.

**Table 6 — Hypothetical v3 Financial Scenarios**

| Scenario                           | UEs avoided | UER savings | False alerts | FA cost   | Net savings | $\Delta$ vs v0 |
|------------------------------------|-------------|-------------|--------------|-----------|-------------|----------------|
| v3 ( $R_C = 0.78$ , $P_C = 0.92$ ) | 7.83        | USD 11.75M  | 0.68         | USD 0.20M | USD 11.55M  | –USD 2.27M     |
| v3 ( $R_C = 0.94$ , $P_C = 0.92$ ) | 9.40        | USD 14.10M  | 0.82         | USD 0.25M | USD 13.85M  | ≈ parity       |

## 4. Migration Recommendation — C-MAPSS to N-CMAPSS

### 4.1 The Case for Migration

Two versions of PiNet retraining on C-MAPSS have failed to close the gap with the v0 XGBoost baseline. The evidence weighs against a third C-MAPSS iteration and in favour of migration to the NASA N-CMAPSS real-flight dataset. The argument is architectural, not training-procedural.

Why migration rather than further C-MAPSS iteration. The C-MAPSS simulator generates relatively clean trajectories where a tabular gradient-boosted model with instantaneous features outperforms a deep temporal architecture. N-CMAPSS provides the continuous flight-envelope variation and higher sensor noise where the temporal denoising hypothesis should actually provide a measurable advantage. Two versions of PiNet have failed to beat the C-MAPSS baseline; a third iteration would face diminishing returns.

**Table 7 — Migration Case Summary**

| Argument for migration                                                                               | Supporting evidence                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C-MAPSS is signal-clean enough that tabular instantaneous features dominate a long-memory deep model | XGBoost RMSE 14.85 vs PiNet 28.05; XGBoost Critical recall 0.94 vs PiNet 0.78 on the metric PiNet was designed to optimise                                                                |
| The regime branch earns its keep only where regime variation is rich                                 | FD002 (six regimes) is v2's best subset at $R_C = 0.98$ ; FD001 (single regime) is among the worst                                                                                        |
| The dual-fault weakness is architectural, not training-related                                       | Persists across v1 and v2 despite materially different training configurations                                                                                                            |
| N-CMAPSS provides continuous flight-envelope variation                                               | Unlike C-MAPSS's six discrete regime centroids, N-CMAPSS exhibits genuine altitude, Mach number and throttle resolver angle continuums — a meaningfully harder task for the regime branch |
| N-CMAPSS sensor noise is higher, and fault propagation is more realistic                             | Closer to actual fleet telemetry, where temporal denoising over a thirty-cycle window provides genuine value rather than redundant smoothing                                              |

## 4.2 Recommended v3 Programme Plan

The recommended v3 programme migrates PiNet unchanged in architecture to N-CMAPSS datasets DS01 through DS08. The full v0-through-v2 statistical preprocessing pipeline is refitted on the N-CMAPSS internal training partition under zero-leakage discipline: per-regime normalisation, compressor efficiency parameters, proportional-hazards model coefficients and the Yeo–Johnson transformation are all re-estimated from training data only. Stage B training hyperparameters — learning rate  $3 \times 10^{-4}$ , dropout 0.25, natural distribution sampler, NASA term scaled by 1/50 — serve as the canonical starting point, removing the confounded-change risk that affected v1. The v0 XGBoost baseline is refit on N-CMAPSS and used as the comparator: if it dominates on N-CMAPSS as it does on C-MAPSS, the conclusion becomes architectural rather than dataset-specific. The v2 calibration framework is re-applied but only after verifying exchangeability between the calibration and test distributions; if the truncation structure that broke v2 coverage persists, a distribution-aware conformal alternative is deployed instead.

## 5. Conclusion

The FusionCore programme delivers two findings of durable operational value. The first is the v0 XGBoost predictive maintenance baseline: safety-gate compliant, financially positive at all fleet sizes evaluated, and ready for operational deployment. The second is a characterised and calibrated PiNet hybrid architecture with a deployment-ready inference wrapper: operationally subordinate to v0 on C-MAPSS but architecturally appropriate for the harder N-CMAPSS problem, with a clearly documented failure mode and a tractable recovery path.

The PiNet hypothesis — that physics-informed temporal modelling supersedes a tabular baseline on turbofan predictive maintenance data — is not validated on C-MAPSS. It is not refuted in general. The architectural diagnosis from four subsets of v2 evaluation data is precise: PiNet's regime branch is effective, and its temporal branch is capable, but both require richer input variation than C-MAPSS's controlled simulation provides. N-CMAPSS, where operational-condition variation is continuous and sensor noise is realistic, is where the hypothesis should next be tested.

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**v0 XGBoost: Operational baseline – ships now**

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**v1 / v2 PiNet: Remain in laboratory – pending v3 N-CMAPSS evaluation**

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## Appendix A — Programme Pain Points: Resolved and Unresolved

### A.1 Resolved

**Table A.1 — Resolved Items**

| Item                                             | Resolution                                                                                                                                                                                                                                        |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v1 Stage A failure mechanism                     | Four-factor decomposition confirmed: excessive learning rate, insufficient temporal regularisation, sampler distribution mismatch, and regression gradient dominance. All four addressed in v2 Stage B, recovering 0.36 units of Critical recall. |
| Distribution-free RUL uncertainty quantification | Split conformal infrastructure operates correctly on in-distribution (validation) data. Exact calibration-slice coverage confirmed. Pluggable to any future model with the same residual interface.                                               |
| Risk-band probability calibration                | Temperature scaling reduces Expected Calibration Error by 71% on validation data. Calibrated probabilities exposed in the production inference wrapper.                                                                                           |
| Deployable inference wrapper                     | The production wrapper composes PiNet, conformal half-widths and temperature scalar into a single object with structured output fields. TorchScript export path included for language-agnostic deployment.                                        |
| Gate placement design lesson                     | Programme lesson formalised: acceptance gates must not be placed on the partition the candidate model was tuned on. All v3 primary gates should be specified against an independent evaluation distribution.                                      |

### A.2 Unresolved

**Table A.2 — Unresolved Items and Proposed v3 Owners**

| Item                                               | Status      | Proposed v3 action                                                                                               |
|----------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------|
| Operational parity with v0 XGBoost                 | Open        | Re-test on N-CMAPSS; if tabular dominance persists, the conclusion becomes architectural not dataset-specific    |
| Conformal coverage on truncated test distributions | Open        | Adopt distribution-aware conformal alternative; evaluate exchangeability before deployment                       |
| Dual-fault fault-mode disambiguation               | Open        | Introduce fault-mode head or richer physics-derived fault-mode features upstream of temporal branch              |
| Full from-raw telemetry inference path             | Open        | v3 deployment task: run the full feature engineering pipeline on raw engine telemetry at inference time          |
| Distribution-shift detection at inference time     | Open        | Add residual-distribution monitoring in the wrapper to alert when incoming data diverges from calibration cohort |
| N-CMAPSS evaluation                                | Not started | Primary v3 programme objective                                                                                   |

## Appendix B — Financial Model Methodology

The financial model in §3 applies a two-component cost structure to each model's Critical-band recall and precision. The primary parameter assumptions are documented in Table B.1. All financial projections are illustrative, derived from published airline maintenance-economics benchmarks, and should be treated as directional rather than contractual estimates.

**Table B.1 — Financial Model Parameters**

| Parameter                                                 | Value                    | Basis                                                                                     |
|-----------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------|
| Cost per Unscheduled Engine Removal (UER)                 | USD 1.5 million          | Industry benchmark: AOG-day rate × average days-out-of-service for an unscheduled removal |
| Cost per false alert                                      | USD 0.30 million         | USD 0.05–0.15M per AOG-day × 2–3 days, mid-range                                          |
| Estimated UERs avoidable per 100-aircraft fleet per annum | ≈ 10                     | Proportional to historical UER incidence rates from RCM literature                        |
| False-alert base rate                                     | Derived from $(1 - P_c)$ | Scaled by the model's positive-prediction rate and fleet size                             |
| UERs avoided                                              | $R_c \times N_{U^U}$     | Linear in Critical recall                                                                 |

The model assumes that every engine genuinely in the Critical band that the model correctly identifies (true positive) translates to one avoided UER. False positives (Healthy or Warning engines predicted as Critical) generate false alerts, each incurring the quoted AOG-day cost. True negatives and false negatives are handled implicitly through the recall and precision terms. The model does not account for additional costs such as parts inventory positioning, maintenance capacity planning, or route disruption beyond the immediate AOG event.